

DERIVATIVES PRICING

A COMPUTATIONAL & THEORETICAL FRAMEWORK

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Part I

Mathematical Foundations

1 Measure Theory

2 Probability

3 Convergence

4 Conditioning

5 Martingales

6 Wiener and Brownian Motion Processes

We defined the Wiener process as a Gaussian process as follows:

Definition 6.1: Wiener Process

The *Wiener process* $(W_t, t \geq 0)$ is a zero-mean Gaussian process with continuous sample paths and covariance function

$$\gamma_{s,t} = \min\{s, t\} =: s \wedge t, \quad s, t \geq 0.$$

A Wiener process is a Lévy process with *continuous* sample paths, as shown in the following result, which can also serve as an equivalent definition of the Wiener process.

Theorem 6.1: Wiener Process as a Lévy Process

$W = (W_t)_{t \geq 0}$ is a Wiener process if and only if the following three properties hold:

- (1) (*Independent increments*): For any $t_1 < t_2 \leq t_3 < t_4$, the increments $W_{t_4} - W_{t_3}$ and $W_{t_2} - W_{t_1}$ are independent.
- (2) (*Gaussian stationarity*): For all $t, u \geq 0$, $W_{t+u} - W_t \sim \mathcal{N}(0, u)$.
- (3) (*Continuity of paths*): W has continuous paths, with $W_0 = 0$.

The following algorithm, [algorithm 1](#) uses the same ideas as in the previous proof to simulate the Wiener process at specific time points $t_1, \dots, t_n$

Algorithm 1 Simulating a Wiener Process

- 1: Let $0 = t_0 < t_1 < \dots < t_n$ be the times at which the process needs to be simulated.
 - 2: Simulate $Z_1, \dots, Z_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$.
 - 3: Return $W_0 := 0$ and $W_{t_k} := \sum_{i=1}^k Z_i \sqrt{t_i - t_{i-1}}$, $k = 1, \dots, n$.
-

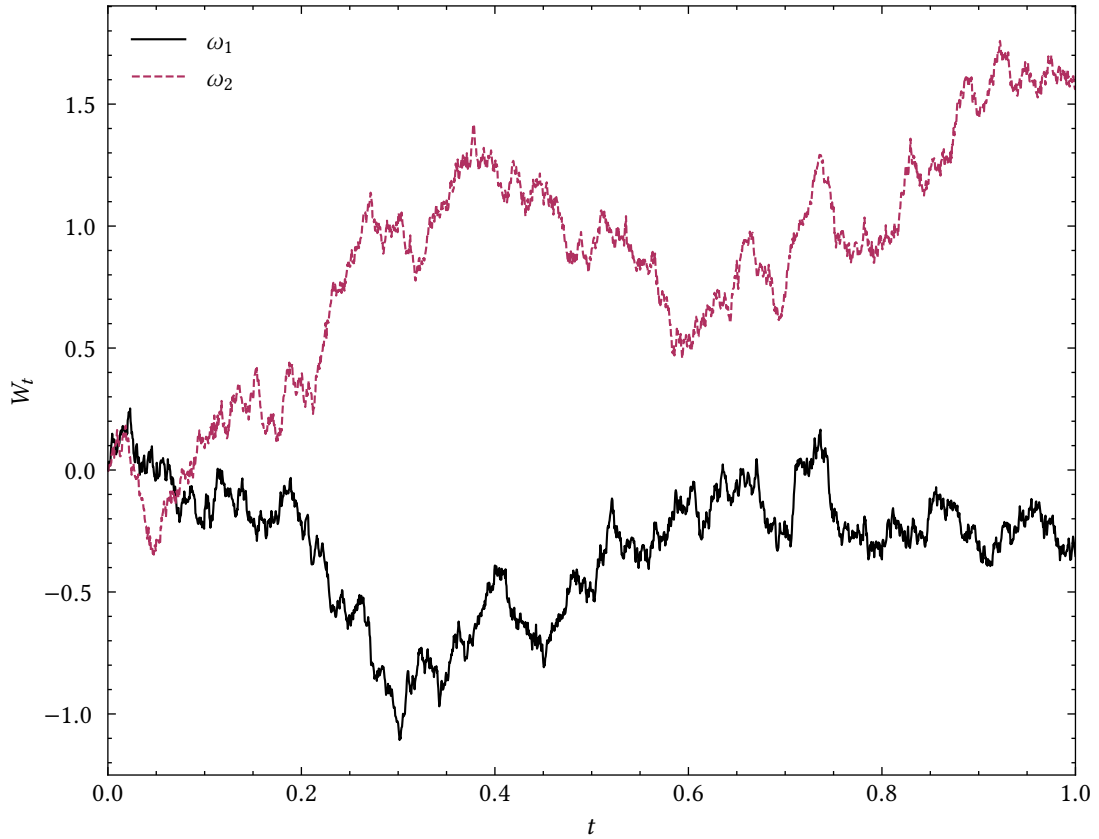


Figure 1: Two realisations of the Wiener process on the interval $[0, 1]$.

Definition 6.2: Brownian Motion

A stochastic process $(B_t, t \geq 0)$ satisfying

$$B_t = B_0 + \mu t + \sigma W_t, \quad t \geq 0,$$

where (W_t) is a Wiener process independent of B_0 , is called a *Brownian motion with drift μ and diffusion coefficient σ* .

A standard Brownian motion is one where $\mu = 0$ and $\sigma = 1$. The only difference between a Wiener process and a Brownian motion is its (random) starting position.

Algorithm 2 Simulating a Brownian Motion

- 1: Simulate the starting position B_0 .
 - 2: Simulate outcomes $W_{t_1}, \dots, W_{t_n}$ of a Wiener process at times $t_1, \dots, t_n$.
 - 3: Return $B_{t_i} := B_0 + \mu t_i + \sigma W_{t_i}$, $i = 1, \dots, n$ as the outcomes of the Brownian motion at times $t_1, \dots, t_n$.
-

Multidimensional Brownian motions are likewise obtained from an affine transformation $B_t = B_0 + \mu t + \sigma W_t$ of a multidimensional Wiener process, where σ is a matrix.

Definition 6.3: Multidimensional Wiener Process

Let $(W_{t,i}, t \geq 0)$, $i = 1, \dots, d$ be independent Wiener processes and let $W_t := [W_{t,1}, \dots, W_{t,d}]^\top$. The process $(W_t, t \geq 0)$ is called a *d-dimensional Wiener process*.

If (W_t) is a d -dimensional Wiener process, then W_t has a $\mathcal{N}(\mathbf{0}, t\mathbf{I}_d)$ multivariate normal distribution, where $\mathbf{I}_d$ denotes the d -dimensional identity matrix.

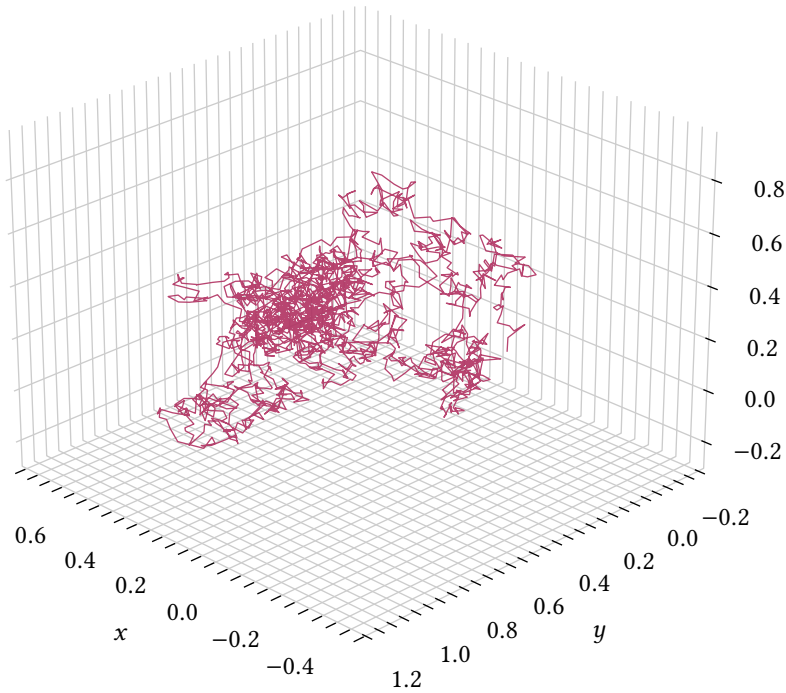


Figure 2: Three-dimensional Wiener process.

Theorem 6.2: Time-reversal and Time-shifting

If $(W_s, s \in [0, t])$ is a Wiener process on $[0, t]$, then so is the time-reversed process $(\widehat{W}_s, s \in [0, t])$ defined by $\widehat{W}_s := W_{t-s} - W_t$. Similarly, if $(W_s, s \geq 0)$ is a Wiener process, then for any $t \geq 0$ the process $(\widetilde{W}_s, s \geq 0)$ defined by $\widetilde{W}_s := W_{t+s} - W_t$ is also a Wiener process.

Theorem 6.3: Scaling

If (W_t) is a Wiener process, then so is (X_t) , with $X_t := W_{at}/\sqrt{a}$, $t \geq 0$ for any $a > 0$.

Lemma 6.1: Probability Bound on the Supremum

For a Wiener process (W_t) , it holds for every $\varepsilon > 0$ that

$$\mathbb{P} \left(\sup_{0 \leq t \leq 1} |W_t| > \varepsilon \right) \leq 2e^{-\varepsilon^2/2}.$$

Theorem 6.4: Reciprocal Time

If (W_t) is a Wiener process, then so is (X_t) , with $X_t := tW_{1/t}$, $t > 0$, $X_0 := 0$.

Theorem 6.5: Existence of the Wiener Process

There exists a probability space $(\Omega, \mathcal{H}, \mathbb{P})$ and a zero-mean Gaussian process $W := (W_t, t \in [0, 1])$ with covariance function $\mathbb{E}W_s W_t = s \wedge t$ and uniformly continuous sample paths.

Theorem 6.6: Right-continuous Filtration

For the Wiener process, the augmentation $(\overline{\mathcal{F}}_t)$ of the natural filtration is right-continuous; that is,

$$\overline{\mathcal{F}}_t = \overline{\mathcal{F}}_t^+ := \bigcap_{\varepsilon > 0} \overline{\mathcal{F}}_{t+\varepsilon}.$$

Theorem 6.7: Wiener Process is Markov w.r.t $\mathcal{F}^+$

Let W be a Wiener process. For every $t \geq 0$, the process $(W_{t+u} - W_t, u \geq 0)$ is independent of $\mathcal{F}_t^+$. Consequently, W is Markovian with respect to $\mathcal{F}^+$.

Theorem 6.8: Strong Markov Property

For any finite stopping time T with respect to $\mathcal{F}^+$, the process $(W_{T+u} - W_T, u \geq 0)$ is a Wiener process independent of $\mathcal{F}_T^+$.

Theorem 6.9: Reflection Principle

Let T be a stopping time. Then, $(\tilde{W}_t, t \geq 0)$, defined by

$$\tilde{W}_t := W_t \mathbb{1}_{\{t \leq T\}} + (2W_T - W_t) \mathbb{1}_{\{t > T\}}, \quad t \geq 0, \quad (1)$$

is a Wiener process.

Theorem 6.10: Transition Kernel of the Wiener Process

The Wiener process W admits the transition function

$$P_{t,u}(x, A) = \mathbb{P}(W_u \in A \mid W_t = x) = \int_A dy p_{u-t}(y \mid x),$$

where the transition density $p_t(y \mid x)$ is given by

$$p_t(y \mid x) := \frac{1}{\sqrt{2\pi t}} e^{-\frac{(y-x)^2}{2t}}, \quad t \geq 0, \quad x, y \in \mathbb{R}. \quad (2)$$

Theorem 6.11: Exponential Martingale

A continuous process $X := (X_t, t \geq 0)$ is a Wiener process if and only if for each $r \in \mathbb{R}$ the process $S := (S_t)$ defined by

$$S_t := e^{rX_t - \frac{1}{2}r^2 t}, \quad t \geq 0, \quad (3)$$

is a martingale.

Theorem 6.12: Martingales from Functions of a Brownian Motion

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be twice continuously differentiable and let $\mathbf{B}$ be d -dimensional standard Brownian motion. If for all $t > 0$ and $\mathbf{x} \in \mathbb{R}^d$,

$$\mathbb{E}^{\mathbf{x}}|f(\mathbf{B}_t)| < \infty \quad \text{and} \quad \mathbb{E}^{\mathbf{x}} \int_0^t |\Delta f(\mathbf{B}_s)| ds < \infty,$$

then the process $(X_t, t \geq 0)$ defined by

$$X_t := f(\mathbf{B}_t) - \frac{1}{2} \int_0^t \Delta f(\mathbf{B}_s) ds \tag{4}$$

is a martingale.

7 Itô Calculus

7.1 Itô Integral

We will work with a Wiener process W on a complete probability space $(\Omega, \mathcal{H}, \mathbb{P})$ with a right-continuous filtration $\mathcal{F} := (\mathcal{F}_s, s \geq 0)$ and such that W is adapted to $\mathcal{F}$. We also use the following definition.

Definition 7.1: Simple Processes

A process $F := (F_s, s \in [0, t])$ is called *simple* if it is of the form

$$F_s := \xi_0 \mathbb{1}_{\{0\}}(s) + \sum_{k=0}^{n-1} \xi_k \mathbb{1}_{(s_k, s_{k+1}]}(s), \quad s \in [0, t], \quad (5)$$

where $\{s_0, \dots, s_n : 0 = s_0 < \dots < s_n = t\} =: \Pi_n, n \in \mathbb{N}$, is a segmentation of $[0, t]$ and $\xi_k \in \mathcal{F}_{s_k}$ for all k .

Note that the simple process in (5) is right-continuous at $s = 0$ and is left-continuous at $s \in (0, t]$. Figure 3 shows a realisation of a simple process on $[0, 1]$ for the segmentation

$$\{s_0, \dots, s_{10}\} = \{0, 0.07, 0.13, 0.19, 0.41, 0.49, 0.52, 0.67, 0.71, 0.76, 1\}.$$

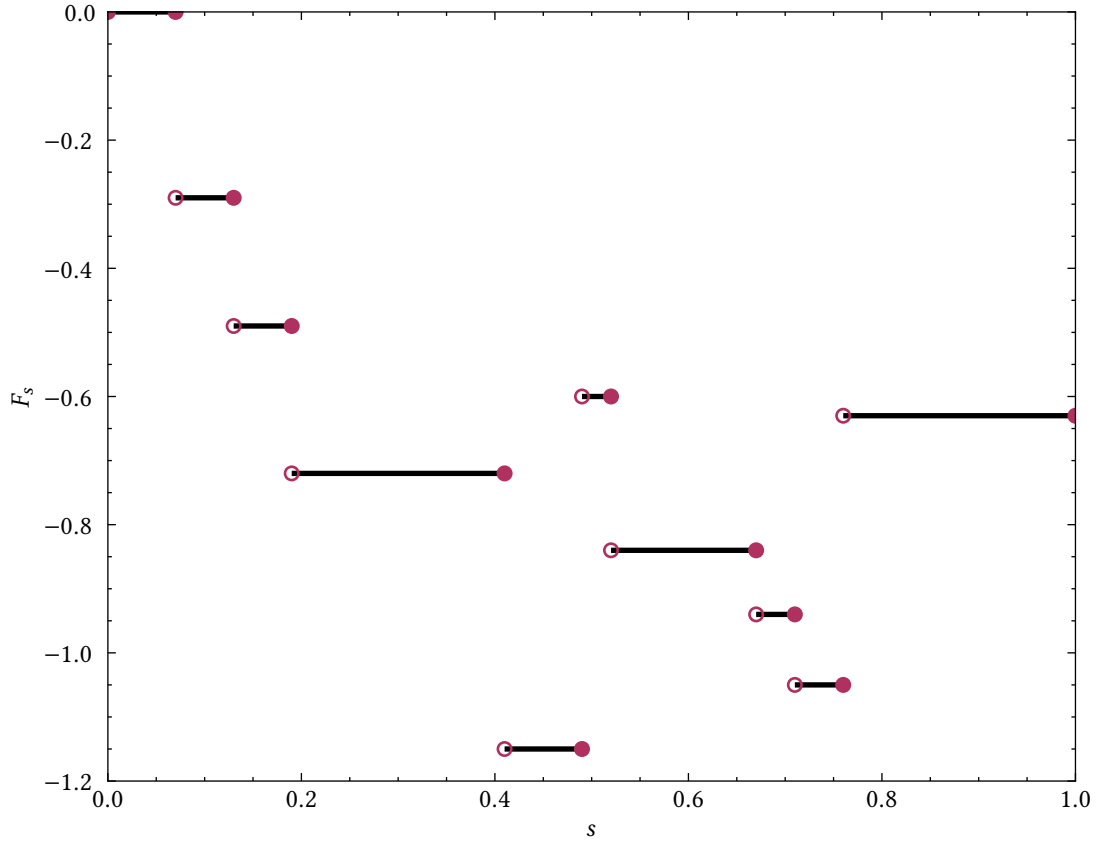


Figure 3: A realisation of a simple process.

Definition 7.2: Itô Integral for Simple Processes

Suppose that F is a simple process of the form eq. (5), with $\int_0^t \mathbb{E} F_s^2 ds < \infty$. Then, the *Itô integral* of F with respect to W over $[0, r]$ is defined as

$$\int_0^r F_s dW_s := \sum_{k=0}^{n-1} \xi_k (W_{s_{k+1} \wedge r} - W_{s_k \wedge r}), \quad r \in [0, t]. \quad (6)$$

Theorem 7.1: Properties of the Itô Integral

Assume that the Itô integral is well-defined. Then, for $\alpha, \beta \in \mathbb{R}$ and $r \in [0, t]$:

- (1) (Linearity): $\int_0^r [\alpha F_s + \beta G_s] dW_s = \alpha \int_0^r F_s dW_s + \beta \int_0^r G_s dW_s$.
- (2) (Zero mean): $\mathbb{E} \int_0^r F_s dW_s = 0$.
- (3) (Isometry): $\mathbb{E} \left[\int_0^r F_s dW_s \times \int_0^r G_s dW_s \right] = \int_0^r \mathbb{E} [F_s G_s] ds$.
- (4) (Martingale): $(\int_0^r F_s dW_s, r \in [0, t])$ is an L^2 -martingale.

Definition 7.3: Canonical Approximation

For any $\mathcal{F}$ -adapted process $F := (F_s, s \in [0, t])$, we define its *canonical approximation* as

$$F_s^{(n)} := F_0 \mathbb{1}_{\{0\}}(s) + \sum_{k=0}^{n-1} F_{s_k} \mathbb{1}_{(s_k, s_{k+1}]}(s), \quad s \in [0, t]. \quad (7)$$

Theorem 7.2: Properties of Left-continuous Adapted Processes

Suppose that $F := (F_s, s \in [0, t])$ is a process on $(\Omega, \mathcal{H}, \mathbb{P})$ that is adapted to the filtration $\mathcal{F}$ and is left-continuous on $(0, t]$. Then, the following hold:

- (1) (Predictable): The process F is $\mathcal{F}$ -predictable (i.e., $\mathcal{F}^P$ -measurable).
- (2) (Progressively Measurable): For each t , the mapping $F : [0, t] \times \Omega \rightarrow \mathbb{R}$ is jointly measurable with respect to $\mathcal{B}_{[0,t]} \otimes \mathcal{F}_t$.
- (3) (Indistinguishable): If $\tilde{F}$ is a modification of F and almost surely left-continuous, then $\mathbb{P}(\cap_{s \leq t} \{F_s = \tilde{F}_s\}) = 1$.

Definition 7.4: The Class of Processes $\mathcal{P}_t$

Let $\mathcal{P}_t$ be the class of stochastic processes $F := (F_s, s \in [0, t])$ that are left-continuous on $(0, t]$, $\mathcal{F}$ -adapted, and satisfy the norm condition:

$$\|F\|_{\mathcal{P}_t}^2 := \mathbb{E} \int_0^t F_s^2 ds < \infty.$$

Theorem 7.3: Approximating the Class $\mathcal{P}_t$ via Simple Processes

Suppose that $F \in \mathcal{P}_t$. Then, there exists a sequence $(F^{(n)})$ of simple processes of the form eq. (5) such that $\text{mesh}(\Pi_n) \rightarrow 0$ implies that $F_s^{(n)} \xrightarrow{\text{a.s.}} F_s$ for all $s \in [0, t]$ and

$$\lim_{n \rightarrow \infty} \|F^{(n)} - F\|_{\mathcal{P}_t} = 0. \quad (8)$$

Proposition 7.1: Deterministic Integrands and Gaussian Processes

Let $f : \mathbb{R}_+ \times \mathbb{R}_+ \rightarrow \mathbb{R}$ be left-continuous in the first argument and such that $\|f(\cdot, t)\|_{\mathcal{P}_t}^2 = \int_0^t f^2(s, t) ds < \infty$. Then, the stochastic integral process

$$Z_t := \int_0^t f(s, t) dW_s, \quad t \geq 0, \quad (9)$$

is a zero-mean Gaussian process with covariance function

$$\text{Cov}(Z_{t_1}, Z_{t_2}) = \int_0^{t_1 \wedge t_2} f(s, t_1) f(s, t_2) ds, \quad t_1, t_2 \geq 0. \quad (10)$$

$$\lim_{m \rightarrow \infty} \sup_{s \in [0, t]} \mathbb{E} F_s^2 \mathbf{1}_{\{F_s^2 > m\}} = 0. \quad (11)$$

Lemma 7.1: Quadratic Variation Integral

Suppose that $F \in \mathcal{P}_t$ satisfies the condition [eq. \(11\)](#). Then, as $\text{mesh}(\Pi_n) \rightarrow 0$:

$$\sum_{k=0}^{n-1} F_{s_k} (W_{s_{k+1}} - W_{s_k})^2 \xrightarrow{L^2} \int_0^t F_s ds. \quad (12)$$

Theorem 7.4: Enlarging the Class $\mathcal{P}_t$ via Localization

Let $F := (F_s, s \in [0, t])$ be an $\mathcal{F}$ -adapted process, left-continuous on $(0, t]$, and such that (7.34) holds. Then, there exists a localizing sequence (T_n) such that the stopped process $(\int_0^{r \wedge T_n} F_s dW_s, r \in [0, t])$ is a square-integrable martingale. In addition, the Itô integral of F with respect to W on $[0, t]$ is defined as the almost sure limit:

$$\int_0^t F_s dW_s := \lim_{n \rightarrow \infty} \int_0^{t \wedge T_n} F_s dW_s. \quad (13)$$

Proposition 7.2: Continuous Modification and Stopping Times

If $F \in \mathcal{P}_t$ and $T \leq t$ is a stopping time with respect to $\mathcal{F}$, then almost surely:

$$\int_0^T F_s dW_s = \int_0^t \mathbf{1}_{[0, T]}(s) F_s dW_s. \quad (14)$$

Theorem 7.5: Continuous Modification of the Itô Integral

Suppose that $F \in \mathcal{P}_t$ and $(F^{(n)})$ is a sequence of simple processes satisfying (7.17). Denote the limit $\int_0^r F_s^{(n)} dW_s =: Z_r^{(n)} \xrightarrow{L^2} Z_r$ as the Itô integral on $[0, r]$ for $r \leq t$. Then, there exists a modification $\tilde{Z}$ of $(Z_r, r \in [0, t])$ such that the mapping $r \mapsto \tilde{Z}_r$ is almost surely continuous. Moreover, this modification is unique up to indistinguishability.

Theorem 7.6: Existence of the Itô Integral with Integrand $F \in \mathcal{P}_t$

Suppose that $F \in \mathcal{P}_t$ and $(F^{(n)})$ is a sequence of simple processes satisfying (7.17). Then, the following limit exists in L^2 norm:

$$\int_0^t F_s dW_s := \lim_{n \rightarrow \infty} \int_0^t F_s^{(n)} dW_s, \quad r \in [0, t], \quad (15)$$

and it defines the Itô integral of F with respect to W on $[0, t]$.

7.2 Itô Calculus**Theorem 7.7: Itô Formula for the Wiener Process**

Let W be a Wiener process and $f : \mathbb{R} \rightarrow \mathbb{R}$ be twice continuously differentiable with derivatives f' and f'' . Then, almost surely

$$f(W_t) - f(W_0) = \int_0^t f'(W_s) dW_s + \frac{1}{2} \int_0^t f''(W_s) ds.$$

Definition 7.5: Integral With Respect To An Itô Process

Let X be an $\text{Itô}(\mu, \sigma)$ process and F a left-continuous $\mathcal{F}$ -adapted process such that almost surely $\int_0^t (|F_s \mu_s| + (F_s \sigma_s)^2) ds < \infty$ for all t . Then, the integral of F with respect to X is defined as the $\text{Itô}(F\mu, F\sigma)$ process with:

$$\int_0^t F_s dX_s := \int_0^t F_s \mu_s ds + \int_0^t F_s \sigma_s dW_s. \quad (16)$$

Definition 7.6: Quadratic Variation On $[0, t]$

Let $\Pi_n := \{s_k, k = 0, \dots, n\}$ be a segmentation of $[0, t]$ and let $(X^{(n)})$ be a sequence of simple processes with almost sure limit $X_s := \lim_n X_s^{(n)}$ for all $s \in [0, t]$ as $\text{mesh}(\Pi_n) \rightarrow 0$. The *quadratic variation* of $X := (X_s, s \in [0, t])$ over $[0, t]$, denoted as $\langle X \rangle_t$, is defined as the limit in probability of

$$\langle X^{(n)} \rangle_t := \sum_{k=0}^{n-1} (X_{s_{k+1}}^{(n)} - X_{s_k}^{(n)})^2. \quad (17)$$

Theorem 7.8: Quadratic Variation Of An Itô Process

If X is an Itô(μ, σ) process, then

$$\langle X^{(n)} \rangle_t \xrightarrow{L^2} \langle X \rangle_t = \int_0^t \sigma_s^2 ds. \quad (18)$$

Furthermore, if $F \in \mathcal{P}_t$ satisfies the condition (7.19) and $\text{mesh}(\Pi_n) \rightarrow 0$, then:

$$\sum_{k=0}^{n-1} F_{s_k} (X_{s_{k+1}} - X_{s_k})^2 \xrightarrow{L^2} \int_0^t F_s \sigma_s^2 ds. \quad (19)$$

Theorem 7.9: Itô's Formula For Itô Processes

Let X be an Itô process and let f be twice continuously differentiable. Then,

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) d\langle X \rangle_s. \quad (20)$$

Definition 7.7: Covariation Of Two Itô Processes

Let X and Y be two Itô processes adapted to the same filtration $\mathcal{F}$. Then, the *covariation* between X and Y is defined as the quantity:

$$\langle X, Y \rangle_t := \frac{\langle X + Y \rangle_t - \langle X \rangle_t - \langle Y \rangle_t}{2}. \quad (21)$$

Theorem 7.10: Itô's Formula In $\mathbb{R}^d$

Let (X_t) be a d -dimensional Itô process, and let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be twice continuously differentiable in all variables. Then,

$$df(X_t) = \sum_{i=1}^d \partial_i f(X_t) dX_{t,i} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \partial_{ij} f(X_t) d\langle X_{\cdot,i}, X_{\cdot,j} \rangle_t. \quad (22)$$

Theorem 7.11: Itô Processes And Martingales

An Itô process in $\mathbb{R}^d$ is a local martingale if and only if its drift is 0.

7.3 Stochastic Differential Equations

Stochastic differential equations are based on the same principle as ordinary differential equations, relating an unknown function to its derivatives, but with the additional feature that part of the unknown function is driven by randomness.

Definition 7.8: Stochastic Differential Equation

A *stochastic differential equation* (SDE) for a stochastic process $(X_t, t \geq 0)$ is an expression of the form

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t, \quad (23)$$

where $(W_t, t \geq 0)$ is a Wiener process and a and b are deterministic functions, called the *drift* and *diffusion* functions. The resulting process $(X_t, t \geq 0)$ is called an *(Itô) diffusion*.

Consider the autonomous SDE

$$dX_t = a(X_t)dt + b(X_t)dW_t. \quad (24)$$

As already mentioned, the meaning of this SDE is that X is a stochastic process that satisfies the integral equation

$$X_t = X_0 + \int_0^t a(X_s)ds + \int_0^t b(X_s)dW_s. \quad (25)$$

However, it is not clear that such a process indeed exists. We show in this section that under mild conditions on a and b we can obtain a unique solution X of the SDE via a method of successive approximations, similar to *Picard iteration* for ordinary differential equations.

The condition that is placed on a and b is that they be *Lipschitz continuous*; that is, for all $x \in \mathbb{R}$ there exists a constant c such that

$$|a(y) - a(x)| + |b(y) - b(x)| \leq c|y - x|. \quad (26)$$

The successive approximation idea is very simple. Suppose that $X_0 = x$. Define $X_t^{(0)} := x$ for all t , and let

$$X_t^{(n+1)} := x + \int_0^t a(X_s^{(n)}) ds + \int_0^t b(X_s^{(n)}) dW_s \quad (27)$$

for $n \in \mathbb{N}$. The following is the main existence result for SDEs:

Theorem 7.12: Existence of SDE Solutions

For any $t \in [0, v]$, the sequence $(X^{(n)})$ in (27) converges almost surely in the uniform norm on $[0, t]$ to a process X that satisfies the integral equation (??).

Lemma 7.2: Integral Bound

Let $X^{(n)}, n = 1, 2, \dots$ be defined in (27), with $X^{(0)} = x$. Then, for every $v > 0$ there exists a constant α such that

$$\mathbb{E} \sup_{s \leq t} \left(X_s^{(n+1)} - X_s^{(n)} \right)^2 \leq \alpha \|X^{(n)} - X^{(n-1)}\|_{\mathcal{P}_t}^2, \quad t \leq v.$$

Definition 7.9: Itô Stratonovich Integral

Let X and Y be Itô processes such that $\int_0^t Y_s dX_s$ and $\int_0^t X_s dY_s$ are well-defined stochastic integrals. Then, the *Itô–Stratonovich integral* of Y with respect to X is defined as

$$\int_0^t Y_s \circ dX_s := \int_0^t Y_s dX_s + \frac{1}{2} \langle X, Y \rangle_t. \quad (28)$$

Proposition 7.3: Itô Stratonovich Formula

Let (X_t) be a d -dimensional Itô process, and let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be three times continuously differentiable in all variables. Then,

$$df(X_t) = \sum_{i=1}^d \partial_i f(X_t) \circ dX_{t,i}. \quad (29)$$

$$\widetilde{\mathbb{P}}^t(A) := \mathbb{E} M_t \mathbb{1}_A, \quad A \in \mathcal{F}_t. \quad (30)$$

$$M_t = \exp \left(\int_0^t \mu_s dW_s - \frac{1}{2} \int_0^t \mu_s^2 ds \right) = e^{Z_t - \frac{1}{2} \langle Z \rangle_t}, \quad t \geq 0, \quad (31)$$

$$\lim_{\varepsilon \downarrow 0} \varepsilon \ln \mathbb{E} \exp((1 - \varepsilon)\langle Z \rangle_t / 2) = 0. \quad (32)$$

Theorem 7.13: Girsanov's Change of Measure

Suppose that μ satisfies the Krylov condition [eq. \(32\)](#) and M is the exponential martingale [eq. \(31\)](#). Then, under the change of measure [eq. \(30\)](#) induced by M , the Itô process

$$\tilde{W}_t := - \int_0^t \mu_s ds + W_t, \quad t \geq 0$$

is a Wiener process.

GBM.py

```

1 N, T = 1000, 1.0
2 mu, sigma = 0.5, 0.2
3 dt = T / N
4 t = np.linspace(0, T, N + 1)
5
6 rng = np.random.default_rng(4)
7 dW = rng.normal(0, np.sqrt(dt), N)
8 W = np.r_[0, np.cumsum(dW)]
9
10 drift = (mu - 0.5 * sigma**2) * t
11 diffusion = sigma * W
12 S_t = np.exp(drift + diffusion)
13 E_St = np.exp(mu * t)

```

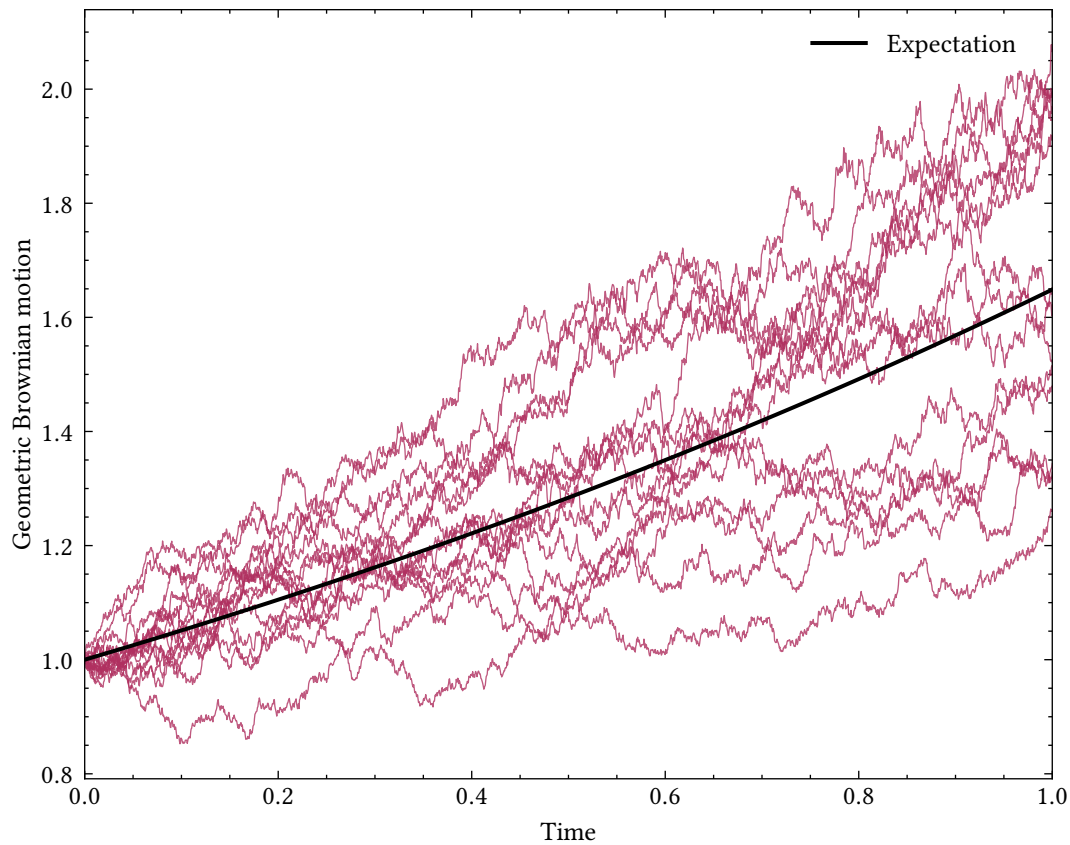


Figure 4: Geometric Brownian motion sample path and expectation for $\mu = 0.5$, $\sigma = 0.2$ and $S_0 = 1$.

Part II

Jump Processes

8 Levy Processes

Consider now a jump-diffusion process

$$X_t = \sigma W_t + \mu t + J_t = X^c(t) + J_t, \quad (33)$$

where J is a compound Poisson process and X^c is the continuous part of X :

$$J_t = \sum_{i=1}^{N_t} \Delta X_i, \quad X_t^c = \mu t + \sigma W_t. \quad (34)$$

Define $Y_t = f(X_t)$ where $f \in C^2(\mathbb{R})$ and denote by $T_i, i = 1 \dots N_T$ the jump times of X . On $]T_i, T_{i+1}[$, X evolves according to

$$dX_t = dX_t^c = \sigma dW_t + \mu dt, \quad (35)$$

so by applying the Itô formula in the Brownian case we obtain

$$\begin{aligned} Y_{T_{i+1}-} - Y_{T_i} &= \int_{T_i}^{T_{i+1}-} \frac{\sigma^2}{2} f''(X_t) dt + \int_{T_i}^{T_{i+1}-} f'(X_t) dX_t \\ &= \int_{T_i}^{T_{i+1}-} \left\{ \frac{\sigma^2}{2} f''(X_t) dt + f'(X_t) dX_t^c \right\} \end{aligned} \quad (36)$$

since $dX_t = dX_t^c(t)$ on this interval. If a jump of size ΔX_t occurs then the resulting change in Y_t is given by $f(X_{t-} + \Delta X_t) - f(X_{t-})$. The total change in Y_t can therefore be written as the sum of these two contributions:

$$\begin{aligned} f(X_t) - f(X_0) &= \int_0^t f'(X_s) dX_s^c + \int_0^t \frac{\sigma^2}{2} f''(X_s) ds \\ &\quad + \sum_{0 \leq s \leq t, \Delta X_s \neq 0} [f(X_{s-} + \Delta X_s) - f(X_{s-})]. \end{aligned} \quad (37)$$

Replacing dX_s^c by $dX_s - \Delta X_s$ one obtains an equivalent expression:

$$\begin{aligned} f(X_t) - f(X_0) &= \int_0^t f'(X_{s-}) dX_s + \int_0^t \frac{\sigma^2}{2} f''(X_s) ds \\ &\quad + \sum_{0 \leq s \leq t, \Delta X_s \neq 0} [f(X_{s-} + \Delta X_s) - f(X_{s-}) - \Delta X_s f'(X_{s-})]. \end{aligned} \quad (38)$$



When the number of jumps is finite, this form is equivalent to [eq. \(37\)](#). However, as we will see below, the form [eq. \(38\)](#) is more general, wherein both the stochastic integral and the sum over the jumps in [eq. \(38\)](#) are well-defined for any semi-martingale, even if the number of jumps is infinite, while the sum in [eq. \(37\)](#) may not converge if jumps have infinite variation.

Here we have only used the Itô formula for diffusions, which is of course still valid if σ is replaced by a process $(\sigma_t)_{t \in [0, T]}$ belonging to the class $\mathcal{P}_T$.

Proposition 8.1: Itô Formula for Jump-Diffusion Processes

Let X be a diffusion process with jumps, defined as the sum of a drift term, a Brownian stochastic integral and a compound Poisson process:

$$X_t = X_0 + \int_0^t b_s ds + \int_0^t \sigma_s dW_s + \sum_{i=1}^{N_t} \Delta X_i,$$

where b_t and σ_t are continuous adapted processes with

$$\mathbb{E} \left[\int_0^T \sigma_t^2 dt \right] < \infty.$$

Then, for any $C^{1,2}$ function $f : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$, the process $Y_t = f(t, X_t)$ can be represented as

$$\begin{aligned} f(t, X_t) - f(0, X_0) &= \int_0^t \left[\frac{\partial f}{\partial s}(s, X_s) + \frac{\partial f}{\partial x}(s, X_s) b_s \right] ds \\ &\quad + \frac{1}{2} \int_0^t \sigma_s^2 \frac{\partial^2 f}{\partial x^2}(s, X_s) ds + \int_0^t \frac{\partial f}{\partial x}(s, X_s) \sigma_s dW_s \\ &\quad + \sum_{\{i \geq 1, T_i \leq t\}} [f(X_{T_i-} + \Delta X_i) - f(X_{T_i-})]. \end{aligned}$$

Expressed in differential notation

$$\begin{aligned} dY_t &= \frac{\partial f}{\partial t}(t, X_t) dt + b_t \frac{\partial f}{\partial x}(t, X_t) dt + \frac{\sigma_t^2}{2} \frac{\partial^2 f}{\partial x^2}(t, X_t) dt \\ &\quad + \frac{\partial f}{\partial x}(t, X_t) \sigma_t dW_t + [f(X_{t-} + \Delta X_t) - f(X_{t-})]. \end{aligned}$$

Definition 8.1: Infinite Divisibility

A probability distribution F on $\mathbb{R}^d$ is said to be *infinitely divisible* if for any integer $n \geq 2$, there exist n i.i.d. random variables $Y_1, \dots, Y_n$ such that $Y_1 + \dots + Y_n$ has distribution F .

Proposition 8.2: Infinite Divisibility and Lévy Processes

Let $X := (X_t)_{t \geq 0}$ be a Lévy process. Then for every t , X_t has an infinitely divisible distribution. Conversely, if F is an infinitely divisible distribution, there exists a Lévy process X such that the distribution of X_1 is given by F .

Proposition 8.3: Characteristic function of a Lévy process

Let $(X_t)_{t \geq 0}$ be a Lévy process on $\mathbb{R}^d$. There exists a continuous function $\psi : \mathbb{R}^d \mapsto \mathbb{R}$ called the characteristic exponent of X , such that:

$$E \left[e^{iz \cdot X_t} \right] = e^{t\psi(z)}, \quad z \in \mathbb{R}^d. \quad (39)$$

9 Integro-differential equations and numerical methods

An approximate answer to the right problem is worth a good deal more than an exact answer to an approximate problem.

John Tukey

Part III

Stochastic Volatility

10 Heston Model

The Heston model is a first-generation model; it owes its popularity to the fact that, being an affine model, the Laplace transform of its moment-generating function for $\ln S$ is analytically known.

Part IV

Risk-Neutral Valuation

11 Assets, Portfolios and Arbitrage

Encyclopedia of Quantitative Finance Encyclopedia of Financial Models

11.1 Portfolio Allocation and Short Selling

We will use the following notation. An element $\bar{x}$ of $\mathbb{R}^{d+1}$ is a vector

$$\bar{x} = \left(x^{(0)}, x^{(1)}, \dots, x^{(d)} \right) \tag{40}$$

made of $d + 1$ components. The scalar product $\bar{x} \cdot \bar{y}$ of two vectors $\bar{x}, \bar{y} \in \mathbb{R}^{d+1}$ is defined by

$$\langle \bar{x}, \bar{y} \rangle := x^{(0)}y^{(0)} + x^{(1)}y^{(1)} + \cdots + x^{(d)}y^{(d)}. \quad (41)$$

The vector

$$\bar{S}_0 = (S_0^{(0)}, S_0^{(1)}, \dots, S_0^{(d)}) \quad (42)$$

denotes the prices at time $t = 0$ of $d + 1$ assets. Namely, $S_0^{(i)} > 0$ is the price at time $t = 0$ of asset $n^o i = 0, 1, \dots, d$.

The asset values $S_1^{(i)} > 0$ of assets No $i = 0, 1, \dots, d$ at time $t = 1$ are represented by the vector

$$\bar{S}_1 = (S_1^{(0)}, S_1^{(1)}, \dots, S_1^{(d)}), \quad (43)$$

whose components $(S_1^{(1)}, \dots, S_1^{(d)})$ are random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

In addition, we will assume that asset $n^o 0$ is a risk-less asset (of savings account type) that yields an interest rate $r > 0$, *i.e.* we have

$$S_1^{(0)} = (1 + r)S_0^{(0)}. \quad (44)$$

Definition 11.1: Portfolio

A portfolio *based on the assets* $0, 1, \dots, d$ is a vector

$$\bar{\xi} = (\xi^{(0)}, \xi^{(1)}, \dots, \xi^{(d)}) \in \mathbb{R}^{d+1}, \quad (45)$$

in which $\xi^{(i)}$ represents the (possibly fractional) quantity of asset $n^o i$ owned by an investor, $i = 0, 1, \dots, d$.

The *price* of such a portfolio, or the cost of the corresponding investment, is given by

$$\bar{\xi} \cdot \bar{S}_0 = \sum_{i=0}^d \xi^{(i)} S_0^{(i)} = \xi^{(0)} S_0^{(0)} + \xi^{(1)} S_0^{(1)} + \cdots + \xi^{(d)} S_0^{(d)} \quad (46)$$

at time $t = 0$. At time $t = 1$, the *value* of the portfolio has evolved into

$$\bar{\xi} \cdot \bar{S}_1 = \sum_{i=0}^d \xi^{(i)} S_1^{(i)} = \xi^{(0)} S_1^{(0)} + \xi^{(1)} S_1^{(1)} + \dots + \xi^{(d)} S_1^{(d)}. \quad (47)$$

There are various ways to construct a portfolio allocation $(\xi^{(i)})_{i=0,1,\dots,d}$.

- i) If $\xi^{(0)} > 0$, the investor puts the amount $\xi^{(0)} S_0^{(0)} > 0$ on a savings account with interest rate r .
- ii) If $\xi^{(0)} < 0$, the investor borrows the amount $-\xi^{(0)} S_0^{(0)} > 0$ with the same interest rate r .
- iii) For $i = 1, 2, \dots, d$, if $\xi^{(i)} > 0$, the investor purchases a (possibly fractional) quantity $\xi^{(i)} > 0$ of the asset $n^o i$.
- iv) If $\xi^{(i)} < 0$, the investor borrows a quantity $-\xi^{(i)} > 0$ of asset i and sells it to obtain the amount $-\xi^{(i)} S_0^{(i)} > 0$.

In the latter case one says that the investor *short sells* a quantity $-\xi^{(i)} > 0$ of the asset $n^o i$, which lowers the cost of the portfolio.

Definition 11.2: Short Selling Ratio

The *short selling ratio*, or percentage of daily turnover activity related to short selling, is defined as the ratio of the number of daily short sold shares divided by daily volume.

Profits are typically generated by first buying at a low price and then selling at a higher price. Short sellers apply the same rule, but in reverse chronological order: first sell high and then buy low, if possible, by following the procedure below.

- (1) Borrow the asset $n^o i$.
- (2) At time $t = 0$, sell the asset $n^o i$ on the market at the price $S_0^{(i)}$ and invest the amount $S_0^{(i)}$ at the interest rate $r > 0$.
- (3) Buy back the asset $n^o i$ at time $t = 1$ at the price $S_1^{(i)}$, with hopefully $S_1^{(i)} < (1 + r)S_0^{(i)}$.
- (4) Return the asset to its owner, with possibly a (small) fee $p > 0$.¹

¹The cost p of short selling will not be taken into account in later calculations.

At the end of the operation, the profit made on share $n^o i$ equals

$$(1+r)S_0^{(i)} - S_1^{(i)} - p > 0, \quad (48)$$

which is positive provided that $S_1^{(i)} < (1+r)S_0^{(i)}$ and $p > 0$ is sufficiently small.

11.2 Arbitrage

11.3 Risk-neutral Probability Measures

11.4 Hedging of Contingent Claims

Definition 11.3: Contingent Claim

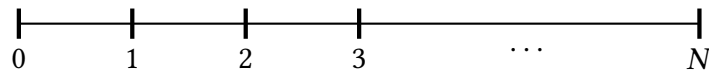
A contingent claim is a financial derivative whose payoff $C : \Omega \rightarrow \mathbb{R}$ is a random variable depending on the realisation of uncertain events.

11.5 Market Completeness

12 Discrete-time Market Model

12.1 Discrete Time Compounding

We work in a finite-horizon discrete-time model indexed by $\{0, 1, \dots, N\}$



We invest a cash flow C each year in an investment plan that carries a constant interest rate r . At the end of the N -th year, the value of the amount C invested at the beginning of year $k = 1, 2, \dots, N$ has turned into $C(1+r)^{N-k+1}$ and the total value of the plan V_N at the

end of the N -th year becomes

$$\begin{aligned}
 V_N &= C \sum_{k=1}^N (1+r)^{N-k+1} \\
 &= C \sum_{k=1}^N (1+r)^k \\
 &= C(1+r) \frac{(1+r)^N - 1}{r},
 \end{aligned} \tag{49}$$

hence the duration N of the plan satisfies

$$N + 1 = \frac{1}{\log(1+r)} \log \left(1 + r + \frac{rV_N}{C} \right). \tag{50}$$

At time $t = 0$ one borrows an amount L over a period of N years at the constant interest rate r per year.

Proposition 12.1: Constant Repayment

Assuming the loan is fully amortized at the beginning of year $N+1$, the constant annual repayment C is uniquely determined by

$$C = \frac{r(1+r)^N L}{(1+r)^N - 1} = \frac{r}{1 - (1+r)^{-N}} L. \tag{51}$$

Proof. Let L_k denote the outstanding principal balance at the beginning of year k , with $L_1 = L$. The evolution of the debt is governed by the linear recurrence relation

$$L_{k+1} = (1+r)L_k - C, \quad \forall k \geq 1. \tag{52}$$

Iteratively expanding this relation yields the value of the debt at the start of any arbitrary year k

$$\begin{aligned}
 L_k &= (1+r)^{k-1} L_1 - C \sum_{j=0}^{k-2} (1+r)^j \\
 &= (1+r)^{k-1} L - C \frac{(1+r)^{k-1} - 1}{r}.
 \end{aligned} \tag{53}$$

The amortization condition requires that the debt is extinguished at the horizon N , such that $L_{N+1} = 0$. Substituting $k = N + 1$ into eq. (53) implies

$$0 = (1 + r)^N L - C \frac{(1 + r)^N - 1}{r}.$$

Rearranging for C yields the result in 12.1. Furthermore, substituting this expression for C back into eq. (53) provides the closed-form outstanding balance

$$L_k = \frac{C}{r} [1 - (1 + r)^{k-1-N}] \equiv \frac{C - (1 + r)^{k-1}(C - rL)}{r}. \quad (54)$$

□

Rearranging the result in 12.1 yields the annuity factor

$$\frac{L}{C} = \frac{1 - (1 + r)^{-N}}{r}, \quad (55)$$

and solving for the duration N implies

$$N = \frac{1}{\log(1 + r)} \log \left(\frac{C}{C - rL} \right) = -\frac{\log(1 - \frac{rL}{C})}{\log(1 + r)}. \quad (56)$$



For N to be finite, we require the condition $C > rL$. This means the annual repayment must exceed the interest accrued on the initial principal; otherwise, the principal balance will never decrease.

Proposition 12.2: Repayment Decomposition

The k -th repayment C decomposes into an interest component rL_k and a principal repayment component $C - rL_k$ given by

$$rL_k = C \left(1 - \frac{1}{(1 + r)^{N-k+1}} \right) = Cr \sum_{l=1}^{N-k+1} (1 + r)^{-l} \quad (57)$$

$$C - rL_k = \frac{C}{(1 + r)^{N-k+1}}, \quad k = 1, 2, \dots, N. \quad (58)$$

Proof. Recall from the proof of 12.1 that the outstanding balance L_k represents the present value of the remaining $N - k + 1$ payments. We express L_k as

$$L_k = \frac{C}{r} \left[1 - (1 + r)^{-(N-k+1)} \right]. \quad (59)$$

Multiplying by r immediately yields the interest component in eq. (57). Consequently, the principal repayment is determined by the residual

$$\begin{aligned} C - rL_k &= C - C \left[1 - (1 + r)^{-(N-k+1)} \right] \\ &= C(1 + r)^{-(N-k+1)}, \end{aligned} \quad (60)$$

which corresponds to eq. (58). $\square$

We verify that the sum of discounted payments at rate r recovers the initial loan amount

$$\sum_{l=1}^N \frac{C}{(1 + r)^l} = C \frac{1 - (1 + r)^{-N}}{r} = L. \quad (61)$$

In particular, the first interest repayment satisfies

$$rL_1 = Cr \sum_{l=1}^N \frac{1}{(1 + r)^l} = C \left(1 - (1 + r)^{-N} \right), \quad (62)$$

and the first principal repayment is

$$C - rL = \frac{C}{(1 + r)^N}. \quad (63)$$

12.2 The Model

Definition 12.1: Numéraire

A *numéraire* is a price process $(X(t))_{t=0}^T$ (a sequence of random variables), which is strictly positive for all $t \in \{0, 1, \dots, T\}$.

For the standard approach the risk-free bank account process is used as numéraire. In some applications, however, it is more convenient to use a security other than the bank account and we therefore just use S_0 without loss of generality.

Definition 12.2: Value of the Portfolio

The value of the portfolio at time t is the scalar product

$$V_\varphi(t) = \varphi(t) \cdot S(t) := \sum_{i=0}^d \varphi_i(t) S_i(t), \quad (t = 1, 2, \dots, T) \text{ and } V_\varphi(0) = \varphi(1) \cdot S(0).$$

The process $V_\varphi(t, \omega)$ is called the *wealth* or *value process* of the trading strategy φ .

The initial wealth $V_\varphi(0)$ is called the *initial investment* or *endowment* of the investor.

Now $\varphi(t) \cdot S(t-1)$ reflects the market value of the portfolio just after it has been established at time $t-1$, whereas $\varphi(t) \cdot S(t)$ is the value just after time t prices are observed, but before changes are made in the portfolio. Hence

$$\varphi(t) \cdot (S(t) - S(t-1)) = \varphi(t) \cdot \Delta S(t)$$

is the change in the market value due to changes in security prices which occur between time $t-1$ and t . This motivates the following definition.

Definition 12.3: Gains Process

The *gains process* G_φ of a trading strategy φ is given by

$$G_\varphi(t) := \sum_{\tau=1}^t \varphi(\tau) \cdot (S(\tau) - S(\tau-1)) = \sum_{\tau=1}^t \varphi(\tau) \cdot \Delta S(\tau), \quad (t = 1, 2, \dots, T).$$

Define $\tilde{S}(t) = (1, \beta(t)S_1(t), \dots, \beta(t)S_d(t))'$, the vector of discounted prices, and consider the *discounted value process*

$$\tilde{V}_\varphi(t) = \beta(t)(\varphi(t) \cdot S(t)) = \varphi(t) \cdot \tilde{S}(t), \quad (t = 1, 2, \dots, T)$$

and the *discounted gains process*

$$\tilde{G}_\varphi(t) := \sum_{\tau=1}^t \varphi(\tau) \cdot (\tilde{S}(\tau) - \tilde{S}(\tau-1)) = \sum_{\tau=1}^t \varphi(\tau) \cdot \Delta \tilde{S}(\tau), \quad (t = 1, 2, \dots, T).$$

Observe that the discounted gains process reflects the gains from trading with assets 1 to d only, which in case of the standard model (a bank account and d stocks) are the risky assets. We will only consider special classes of trading strategies.

Definition 12.4: Self-Financing Strategy

The strategy φ is *self-financing*, $\varphi \in \Phi$, if

$$\varphi(t) \cdot S(t) = \varphi(t+1) \cdot S(t) \quad (t = 1, 2, \dots, T-1). \quad (64)$$

When new prices $S(t)$ are quoted at time t , the investor adjusts his portfolio from $\varphi(t)$ to $\varphi(t+1)$, without bringing in or consuming any wealth. The following result (which is trivial in our current setting, but requires a little argument in continuous time) shows that renormalising security prices (i.e. changing the numéraire) has essentially no economic effects.

Proposition 12.3: Numéraire Invariance

Let $X(t)$ be a numéraire. A trading strategy φ is self-financing with respect to $S(t)$ if and only if φ is self-financing with respect to $X(t)^{-1}S(t)$.

Proof. Since $X(t) > 0$ for all t , scalar multiplication by $X(t)^{-1}$ preserves the equality. For $t = 1, \dots, T-1$:

$$\begin{aligned} \varphi(t) \cdot S(t) &= \varphi(t+1) \cdot S(t) \\ \iff \varphi(t) \cdot (X(t)^{-1}S(t)) &= \varphi(t+1) \cdot (X(t)^{-1}S(t)). \end{aligned}$$

□

Corollary 12.1: Self-Financing Equivalence

A trading strategy φ is self-financing with respect to $S(t)$ if and only if φ is self-financing with respect to $\tilde{S}(t)$.

We now give a characterisation of self-financing strategies in terms of the discounted processes.

Proposition 12.4: Characterisation of Self-Financing Strategies

A trading strategy ϕ belongs to Φ if and only if

$$\tilde{V}_\phi(t) = V_\phi(0) + \tilde{G}_\phi(t), \quad t = 0, 1, \dots, T. \quad (65)$$

Proof. Assume $\phi \in \Phi$. Then, using the defining relation of the gains process, the numéraire invariance theorem, and the fact that $\tilde{S}_0^{(0)} = 1$:

$$\begin{aligned} V_\phi(0) + \tilde{G}_\phi(t) &= \phi_1 \cdot S_0 + \sum_{\tau=1}^t \phi_\tau \cdot (\tilde{S}_\tau - \tilde{S}_{\tau-1}) \\ &= \phi_1 \cdot \tilde{S}_0 + \phi_t \cdot \tilde{S}_t - \sum_{\tau=1}^{t-1} (\phi_{\tau+1} - \phi_\tau) \cdot \tilde{S}_\tau - \phi_1 \cdot \tilde{S}_0 \\ &= \phi_t \cdot \tilde{S}_t \\ &= \tilde{V}_\phi(t). \end{aligned}$$

Note that the summation term vanishes because the strategy is self-financing (i.e., $\phi_{\tau+1} \cdot \tilde{S}_\tau = \phi_\tau \cdot \tilde{S}_\tau$).

Conversely, assume that [eq. \(65\)](#) holds true. By the numéraire invariance theorem, it is sufficient to show the discounted version of the self-financing relation for each $t = 1, 2, \dots, T-1$. Summing up to $t = 2$, [eq. \(65\)](#) yields:

$$\phi_2 \cdot \tilde{S}_2 = \phi_1 \cdot \tilde{S}_0 + \phi_1 \cdot (\tilde{S}_1 - \tilde{S}_0) + \phi_2 \cdot (\tilde{S}_2 - \tilde{S}_1).$$

Subtracting $\phi_2 \cdot \tilde{S}_2$ from both sides simplifies to:

$$0 = \phi_1 \cdot \tilde{S}_1 - \phi_2 \cdot \tilde{S}_1 \implies \phi_2 \cdot \tilde{S}_1 = \phi_1 \cdot \tilde{S}_1.$$

Proceeding similarly—or by induction—we can demonstrate that $\phi_t \cdot \tilde{S}_t = \phi_{t+1} \cdot \tilde{S}_t$ for $t = 2, \dots, T-1$, as required. $\square$

We are allowed to borrow (so $\phi_t^{(0)}$ may be negative) and sell short (so $\phi_t^{(i)}$ may be negative for $i = 1, \dots, d$). Consequently, if we decide on the allocation of the risky assets, the numéraire component adjusts automatically to satisfy the budget constraint.

Proposition 12.5: Predictability of the Numéraire Component

If $(\phi_1(t), \dots, \phi_d(t))'$ is predictable and V_0 is $\mathcal{F}_0$ -measurable, there is a unique predictable process $(\phi_0(t))_{t=1}^T$ such that $\phi = (\phi_0, \phi_1, \dots, \phi_d)'$ is self-financing with initial value of the corresponding portfolio $V_\phi(0) = V_0$.

Proof. If ϕ is self-financing, then by [Proposition 12.4](#),

$$\tilde{V}_\phi(t) = V_0 + \tilde{G}_\phi(t) = V_0 + \sum_{\tau=1}^t \left(\phi_1(\tau) \Delta \tilde{S}_1(\tau) + \dots + \phi_d(\tau) \Delta \tilde{S}_d(\tau) \right).$$

On the other hand,

$$\tilde{V}_\phi(t) = \phi(t) \cdot \tilde{S}(t) = \phi_0(t) + \phi_1(t) \tilde{S}_1(t) + \dots + \phi_d(t) \tilde{S}_d(t).$$

Equate these:

$$\begin{aligned} \phi_0(t) &= V_0 + \sum_{\tau=1}^t \left(\phi_1(\tau) \Delta \tilde{S}_1(\tau) + \dots + \phi_d(\tau) \Delta \tilde{S}_d(\tau) \right) \\ &\quad - \left(\phi_1(t) \tilde{S}_1(t) + \dots + \phi_d(t) \tilde{S}_d(t) \right), \end{aligned}$$

which defines $\phi_0(t)$ uniquely. The terms in $\tilde{S}_i(t)$ are

$$\phi_i(t) \Delta \tilde{S}_i(t) - \phi_i(t) \tilde{S}_i(t) = -\phi_i(t) \tilde{S}_i(t-1),$$

which is $\mathcal{F}_{t-1}$ -measurable. So

$$\begin{aligned}\phi_0(t) = V_0 &+ \sum_{\tau=1}^{t-1} \left(\phi_1(\tau) \Delta \tilde{S}_1(\tau) + \cdots + \phi_d(\tau) \Delta \tilde{S}_d(\tau) \right) \\ &- \left(\phi_1(t) \tilde{S}_1(t-1) + \cdots + \phi_d(t) \tilde{S}_d(t-1) \right),\end{aligned}$$

where as $\phi_1, \dots, \phi_d$ are predictable, all terms on the right-hand side are $\mathcal{F}_{t-1}$ -measurable, so ϕ_0 is predictable. $\square$

12.3 Existence of Equivalent Martingale Measures

Definition 12.5: Arbitrage Opportunity

Let $\tilde{\Phi} \subset \Phi$ be a set of self-financing strategies. A strategy $\varphi \in \tilde{\Phi}$ is called an *arbitrage opportunity* or *arbitrage strategy* with respect to $\tilde{\Phi}$ if $\mathbb{P}\{V_\varphi(0) = 0\} = 1$, and the terminal wealth of φ satisfies

$$\mathbb{P}\{V_\varphi(T) \geq 0\} = 1 \quad \text{and} \quad \mathbb{P}\{V_\varphi(T) > 0\} > 0.$$

An arbitrage opportunity is a self-financing strategy with zero initial value, which produces a non-negative final value with probability one and has a positive probability of yielding a positive final value. Note that arbitrage opportunities are always defined in relation to a specific class of trading strategies.

Definition 12.6: Arbitrage-Free Market

We say that a security market $\mathcal{M}$ is arbitrage-free if there are no arbitrage opportunities in the class Φ of trading strategies. We will allow ourselves to use ‘no-arbitrage’ in place of ‘arbitrage-free’ when convenient.

Lemma 12.1: Global Implies Local No-Arbitrage

If the market model contains no arbitrage opportunities, then for all $t \in \{0, 1, \dots, T-1\}$, for all self-financing trading strategies $\varphi \in \Phi$ and for any $A \in \mathcal{P}_t$, we have

$$(i) \quad \mathbb{P}(\tilde{V}_\varphi(t+1) - \tilde{V}_\varphi(t) \geq 0 \mid A) = 1 \Rightarrow \mathbb{P}(\tilde{V}_\varphi(t+1) - \tilde{V}_\varphi(t) = 0 \mid A) = 1,$$

$$(ii) \quad \mathbb{P}(\tilde{V}_\varphi(t+1) - \tilde{V}_\varphi(t) \leq 0 \mid A) = 1 \Rightarrow \mathbb{P}(\tilde{V}_\varphi(t+1) - \tilde{V}_\varphi(t) = 0 \mid A) = 1.$$

The economic meaning: no arbitrage ‘globally’ implies no arbitrage ‘locally’. Any local trading strategy can be embedded in a global strategy to which the global no-arbitrage condition applies.

Proof. We only prove (i) ((ii) is similar). Fix $t \in \{0, \dots, T-1\}$ and $\varphi \in \Phi$. Suppose $\mathbb{P}(\tilde{V}_\varphi(t+1) - \tilde{V}_\varphi(t) \geq 0 \mid A) = 1$ for some $A \in \mathcal{P}_t$ and define a new trading strategy ψ for all times $u = 1, \dots, T$ as follows: For $u \leq t$: $\psi(u) = 0$ (‘do nothing before time t ’). For $u = t+1$: $\psi(t+1) = 0$ if $\omega \notin A$, and

$$\psi_k(t+1, \omega) = \begin{cases} \varphi_k(t+1, \omega) & \text{if } \omega \in A \text{ and } k \in \{1, \dots, d\}, \\ \varphi_0(t+1, \omega) - \tilde{V}_\varphi(t, \omega) & \text{if } \omega \in A \text{ and } k = 0. \end{cases}$$

(If $\omega \in A$ at time t , follow φ in risky assets, but adjust the numéraire to offset doing nothing when $\omega \notin A$.) For $u > t+1$: $\psi_k(u) = 0$ for $k \in \{1, \dots, d\}$ and

$$\psi_0(u, \omega) = \begin{cases} \tilde{V}_\psi(t+1, \omega) & \text{if } \omega \in A, \\ 0 & \text{if } \omega \notin A. \end{cases}$$

(Invest $\tilde{V}_\psi(t+1)$ in the numéraire if $\omega \in A$, otherwise do nothing.)

We show ψ is self-financing. By construction, ψ is predictable. For $\omega \notin A$, $\psi \equiv 0$, so consider $\omega \in A$. The relevant time is $t+1$. Since $\psi(t) = 0$, $\psi(t) \cdot \tilde{S}(t) = 0$. Now

$$\begin{aligned} \psi(t+1) \cdot \tilde{S}(t) &= (\varphi_0(t+1) - \tilde{V}_\varphi(t))\tilde{S}_0(t) + \sum_{k=1}^d \varphi_k(t+1)\tilde{S}_k(t) \\ &= \sum_{k=0}^d \varphi_k(t+1)\tilde{S}_k(t) - \tilde{V}_\varphi(t) \\ &= \varphi(t+1) \cdot \tilde{S}(t) - \tilde{V}_\varphi(t) = \varphi(t) \cdot \tilde{S}(t) - \tilde{V}_\varphi(t) = 0, \end{aligned}$$

since φ is self-financing. For $u \leq t$, $\psi(u) \cdot \tilde{S}(u) = 0$, hence $\psi(u+1) \cdot \tilde{S}(u) = \psi(u) \cdot \tilde{S}(u)$ for all $u \leq t$. When $u > t+1$ and $\omega \in A$ we only hold the numéraire (discounted value = 1), so

$$\psi(u+1) \cdot \tilde{S}(u) = \tilde{V}_\psi(t+1) = \psi(u) \cdot \tilde{S}(u).$$

Therefore, ψ is self-financing.

We now analyze the value process of ψ . Using $\mathbb{P}(\tilde{V}_\varphi(t+1) - \tilde{V}_\varphi(t) \geq 0 \mid A) = 1$, for all $u \geq t+1$ and $\omega \in A$,

$$\begin{aligned} \tilde{V}_\psi(u) &= \psi(u) \cdot \tilde{S}(u) = \psi(t+1) \cdot \tilde{S}(t+1) \\ &= (\varphi_0(t+1) - \tilde{V}_\varphi(t))\tilde{S}_0(t+1) + \sum_{k=1}^d \varphi_k(t+1)\tilde{S}_k(t+1) \\ &= \sum_{k=0}^d \varphi_k(t+1)\tilde{S}_k(t+1) - \tilde{V}_\varphi(t) = \tilde{V}_\varphi(t+1) - \tilde{V}_\varphi(t) \geq 0. \end{aligned}$$

Since $\tilde{V}_\psi(T) = 0$ on A^c , ψ is self-financing with $\tilde{V}_\psi(0) = 0$ and $\tilde{V}_\psi(T) \geq 0$. Arbitrage-free implies $\tilde{V}_\psi(T) = 0$, hence

$$\begin{aligned} 0 &= \mathbb{P}(\tilde{V}_\psi(T) > 0) = \mathbb{P}(\{\tilde{V}_\psi(T) > 0\} \cap A) \\ &= \mathbb{P}(\tilde{V}_\psi(t+1) - \tilde{V}_\psi(t) > 0 \mid A) \mathbb{P}(A). \end{aligned}$$

Therefore $\mathbb{P}(\tilde{V}_\psi(t+1) - \tilde{V}_\psi(t) = 0 \mid A) = 1$. □

Definition 12.7: Martingale Measure

A probability measure $\mathbb{P}^*$ on $(\Omega, \mathcal{F}_T)$ equivalent to $\mathbb{P}$ is called a *martingale measure* for $\tilde{S}$ if the process $\tilde{S}$ follows a $\mathbb{P}^*$ -martingale with respect to the filtration $\mathcal{F}$. We denote by $\mathcal{P}(\tilde{S})$ the class of equivalent martingale measures.

Proposition 12.6: Wealth Process Martingale

Let $\mathbb{P}^*$ be an equivalent martingale measure ($\mathbb{P}^* \in \mathcal{P}(\tilde{S})$) and $\varphi \in \Phi$ any self-financing strategy. Then the wealth process $\tilde{V}_\varphi(t)$ is a $\mathbb{P}^*$ -martingale with respect to the filtration $\mathcal{F}$.

Proof. By the self-financing property of φ (compare [Proposition 12.4](#) and [eq. \(65\)](#)) and, we have

$$\tilde{V}_\varphi(t) = V_\varphi(0) + \tilde{G}_\varphi(t) \quad (t = 0, 1, \dots, T).$$

So

$$\tilde{V}_\varphi(t+1) - \tilde{V}_\varphi(t) = \tilde{G}_\varphi(t+1) - \tilde{G}_\varphi(t) = \varphi(t+1) \cdot (\tilde{S}(t+1) - \tilde{S}(t)).$$

So for $\varphi \in \Phi$, $\tilde{V}_t(\varphi)$ is the martingale transform of the $\mathbb{P}^*$ martingale $\tilde{S}$ by φ and hence a $\mathbb{P}^*$ martingale itself. $\square$

The next result is the key to further development.

Proposition 12.7: Equivalent Martingale Measure Implies Arbitrage-Free

If an equivalent martingale measure exists - that is, if $\mathcal{P}(\tilde{S}) \neq \emptyset$ - then the market $\mathcal{M}$ is arbitrage-free.

Proof. Assume such a $\mathbb{P}^*$ exists. For any self-financing strategy φ , we have as before

$$\tilde{V}_\varphi(t) = V_\varphi(0) + \sum_{\tau=1}^t \varphi(\tau) \cdot \Delta \tilde{S}(\tau).$$

By Proposition 12.4, $\tilde{S}(t)$ a (vector) $\mathbb{P}^*$ -martingale implies $\tilde{V}_\varphi(t)$ is a $\mathbb{P}^*$ -martingale. So the initial and final $\mathbb{P}^*$ -expectations are the same,

$$\mathbb{E}^*(\tilde{V}_\varphi(T)) = \mathbb{E}^*(\tilde{V}_\varphi(0)).$$

If the strategy is an arbitrage opportunity, its initial value - the right-hand side above - is zero. Therefore the left-hand side $\mathbb{E}^*(\tilde{V}_\varphi(T))$ is zero, but $\tilde{V}_\varphi(T) \geq 0$ (by definition). Also each $\mathbb{P}^*(\{\omega\}) > 0$ (by assumption, each $\mathbb{P}(\{\omega\}) > 0$, so by equivalence each $\mathbb{P}^*(\{\omega\}) > 0$). This and $\tilde{V}_\varphi(T) \geq 0$ force $\tilde{V}_\varphi(T) = 0$. So no arbitrage is possible. $\square$

Proposition 12.8: Arbitrage-Free Implies Existence of EMM

If the market $\mathcal{M}$ is arbitrage-free, then the class $\mathcal{P}(\tilde{S})$ of equivalent martingale measures is non-empty.

Because of the fundamental nature of this result, we will provide two proofs. The first proof is based on our previous observation that the ‘global’ no-arbitrage condition implies also no-arbitrage ‘locally’. We therefore can combine single-period results to prove the

multi-period claim. The second proof uses functional-analytic techniques, i.e. a variant of the *Hahn-Banach theorem*.

Theorem 12.1: No-Arbitrage Theorem

The market $\mathcal{M}$ is arbitrage-free if and only if there exists a probability measure $\mathbb{P}^$ equivalent to $\mathbb{P}$ under which the discounted d -dimensional asset price process $\tilde{S}$ is a $\mathbb{P}^*$ -martingale.*

12.4 Pricing

We now turn to the main underlying question of this text, namely the pricing of contingent claims (i.e. financial derivatives). First, we have to model these financial instruments in our current framework. This is done in the following fashion.

Definition 12.8: Contingent Claim

A contingent claim X with maturity date T is an arbitrary non-negative $\mathcal{F}_T$ -measurable random variable (which is, by the finiteness of the probability space, bounded). We denote the class of all contingent claims by $\mathcal{X}^+$.

A typical example of a contingent claim X is an option on some underlying asset S , then (e.g. for the case of a European call option with maturity date T and strike K) we have a functional relation $X = f(S)$ with some function f (e.g. $X = (S(T) - K)^+$). The general definition allows for more complicated relationships, which are captured by the $\mathcal{F}_T$ -measurability of X (recall that $\mathcal{F}_T$ is typically generated by the process S).

We say that the claim is *attainable* if there exists a *replicating strategy* $\varphi \in \Phi$ such that

$$V_\varphi(T) = X.$$

Therefore, the replicating strategy generates the same cash flow at time T as does X . In a highly efficient security market, we expect the law of one price to hold; that is, for a specified cash flow, there exists only one price at any given instant. Otherwise, arbitrageurs would use the opportunity to cash in a riskless profit. So the no-arbitrage condition implies that for an attainable contingent claim, its time t price must be given by the value of any

replicating strategy (we say the claim is uniquely replicated in that case). This is the basic idea of the arbitrage pricing theory. Clearly, the equivalence of the no-arbitrage condition and the existence of risk-neutral probability measures implies that risk-neutral measures can be used for pricing purposes.

Proposition 12.9: Unique Replication

Suppose the market $\mathcal{M}$ is arbitrage-free. Then any attainable contingent claim X is uniquely replicated in $\mathcal{M}$.

Proof. Suppose there is an attainable contingent claim X and strategies φ and ψ such that

$$V_\varphi(T) = V_\psi(T) = X,$$

but there exists a $\tau < T$ such that

$$V_\varphi(u) = V_\psi(u) \text{ for every } u < \tau \text{ and } V_\varphi(\tau) \neq V_\psi(\tau).$$

Define $A := \{\omega \in \Omega : V_\varphi(\tau, \omega) > V_\psi(\tau, \omega)\}$, then $A \in \mathcal{F}_\tau$ and $\mathbb{P}(A) > 0$ (otherwise just rename the strategies). Define the $\mathcal{F}_\tau$ -measurable random variable $Y := V_\varphi(\tau) - V_\psi(\tau)$ and consider the trading strategy ξ defined by

$$\xi(u) = \begin{cases} \varphi(u) - \psi(u), & u \leq \tau \\ \mathbb{1}_{A^c}(\varphi(u) - \psi(u)) + \mathbb{1}_A(Y\beta(\tau), 0, \dots, 0), & \tau < u \leq T. \end{cases}$$

Then ξ is predictable, and the self-financing condition [Equation \(64\)](#) is clearly true for $t \neq \tau$, and for $t = \tau$ we have using that $\varphi, \psi \in \Phi$

$$\begin{aligned} \xi(\tau) \cdot S(\tau) &= (\varphi(\tau) - \psi(\tau)) \cdot S(\tau) = V_\varphi(\tau) - V_\psi(\tau), \\ \xi(\tau + 1) \cdot S(\tau) &= \mathbb{1}_{A^c}(\varphi(\tau + 1) - \psi(\tau + 1)) \cdot S(\tau) + \mathbb{1}_A Y \beta(\tau) S_0(\tau) \\ &= \mathbb{1}_{A^c}(\varphi(\tau) - \psi(\tau)) \cdot S(\tau) + \mathbb{1}_A (V_\varphi(\tau) - V_\psi(\tau)) \beta(\tau) \beta^{-1}(\tau) \\ &= V_\varphi(\tau) - V_\psi(\tau). \end{aligned}$$

Hence ξ is a self-financing strategy with initial value equal to zero. Furthermore

$$\begin{aligned} V_\xi(T) &= \mathbb{1}_{A^c}(\varphi(T) - \psi(T)) \cdot S(T) + \mathbb{1}_A(Y\beta(\tau), 0, \dots, 0) \cdot S(T) \\ &= \mathbb{1}_A Y\beta(\tau) S_0(T) \geq 0 \end{aligned}$$

and

$$\mathbb{P}\{V_\xi(T) > 0\} = \mathbb{P}\{A\} > 0.$$

Hence, the market $\mathcal{M}$ contains an arbitrage opportunity with respect to the class Φ of self-financing strategies. But this contradicts the assumption that the market $\mathcal{M}$ is arbitrage-free. $\square$

This uniqueness property allows us now to define the important concept of an arbitrage price process.

Definition 12.9: Arbitrage Price Process

Suppose the market $\mathcal{M}$ is arbitrage-free. Let X be any attainable contingent claim with time T maturity. Then the *arbitrage price process* $\pi_X(t)$, $0 \leq t \leq T$ or simply *arbitrage price* of X in $\mathcal{M}$ is given by the value process of any replicating strategy φ for X .

The synthesis of hedging strategies that replicate the outcome of a contingent claim constitutes a fundamental objective in both theoretical and applied finance. Classical valuation frameworks postulate that an option is perfectly replicated by a dynamic self-financing portfolio of the underlying market assets (Black and Scholes, 1973; Merton, 1974). This mechanism enables the construction of a riskless hedge that neutralises stochastic exposure and informs the delta-hedging strategies employed by market participants.

The resulting pricing functional depends strictly on the assumption of non-satiation to preclude arbitrage opportunities and remains invariant to specific investor utility functions. Consequently, the value derived in an actual market must coincide with that of a hypothetical economy populated entirely by risk-neutral agents (Cox and Ross, 1976). This principle enables the valuation of derivatives as the expected discounted payoff under a

specific probability measure, where discounted asset prices form martingales (Harrison and Pliska, 1981).

Proposition 12.10: Risk-Neutral Valuation Formula

The arbitrage price process of any attainable contingent claim X is given by the risk-neutral valuation formula

$$\pi_X(t) = \beta(t)^{-1} \mathbb{E}^*(X\beta(T) \mid \mathcal{F}_t) \quad \forall t = 0, 1, \dots, T, \quad (66)$$

where $\mathbb{E}^$ is the expectation operator with respect to an equivalent martingale measure $\mathbb{P}^*$.*

Proof. Since we assume the market is arbitrage-free there exists (at least) an equivalent martingale measure $\mathbb{P}^*$. By Proposition 12.6 the discounted value process $\tilde{V}_\varphi$ of any self-financing strategy φ is a $\mathbb{P}^*$ -martingale. So for any contingent claim X with maturity T and any replicating trading strategy $\varphi \in \Phi$ we have for each $t = 0, 1, \dots, T$

$$\begin{aligned} \pi_X(t) &= V_\varphi(t) = \beta(t)^{-1} \tilde{V}_\varphi(t) \\ &= \beta(t)^{-1} \mathbb{E}^*(\tilde{V}_\varphi(T) \mid \mathcal{F}_t) \quad (\text{as } \tilde{V}_\varphi(t) \text{ is a } \mathbb{P}^*\text{-martingale}) \\ &= \beta(t)^{-1} \mathbb{E}^*(\beta(T)V_\varphi(T) \mid \mathcal{F}_t) \quad (\text{undoing the discounting}) \\ &= \beta(t)^{-1} \mathbb{E}^*(\beta(T)X \mid \mathcal{F}_t) \quad (\text{as } \varphi \text{ is a replicating strategy for } X). \end{aligned} \quad (67)$$

□

12.5 Market Completeness and EMMs

The last section made clear that attainable contingent claims can be priced using an equivalent martingale measure. In this section we will discuss the question of the circumstances under which all contingent claims are attainable. This would be a very desirable property of the market $\mathcal{M}$, because we would then have solved the pricing question (at least for contingent claims) completely. Since contingent claims are merely non-negative $\mathcal{F}_T$ -measurable random variables in our setting, it should be no surprise that we can give a criterion in terms of probability measures. We start with:

Definition 12.10: Market Completeness

A market $\mathcal{M}$ is complete if every contingent claim is attainable, i.e. for every non-negative $\mathcal{F}_T$ -measurable random variable $X \in \mathcal{X}^+$ there exists a replicating self-financing strategy $\varphi \in \Phi$ such that $V_\varphi(T) = X$.

In the case of an arbitrage-free market $\mathcal{M}$ one can even insist on replicating contingent claims by an admissible strategy $\varphi \in \Phi_a$. Indeed, if φ is self-financing and $\mathbb{P}^*$ is an equivalent martingale measure under which discounted prices $\tilde{S}$ are $\mathbb{P}^*$ -martingales (such $\mathbb{P}^*$ exist since $\mathcal{M}$ is arbitrage-free and we can hence use the no-arbitrage theorem, [Theorem 12.1](#)) $\tilde{V}_\varphi(t)$ is also a $\mathbb{P}^*$ -martingale, being the martingale transform of the martingale $\tilde{S}$ by φ (see [Proposition 12.6](#)). So

$$\tilde{V}_\varphi(t) = \mathbb{E}^*(\tilde{V}_\varphi(T) | \mathcal{F}_t) \quad (t = 0, 1, \dots, T). \quad (68)$$

If φ replicates X , $V_\varphi(T) = X \geq 0$, so discounting, $\tilde{V}_\varphi(T) \geq 0$, so the above equation gives $\tilde{V}_\varphi(t) \geq 0$ for each t . Thus all the values at each time t are non-negative, not just the final value at time T , so φ is admissible.

Theorem 12.2: Completeness Theorem

An arbitrage-free market $\mathcal{M}$ is complete if and only if there exists a unique probability measure $\mathbb{P}^$ equivalent to $\mathbb{P}$ under which discounted asset prices are martingales.*

Proof. ' $\Rightarrow$ ': Assume that the arbitrage-free market $\mathcal{M}$ is complete. Then for any $\mathcal{F}_T$ -measurable random variable $X \geq 0$ (contingent claim), there exists an admissible (so self-financing) strategy φ replicating X : $X = V_\varphi(T)$. As φ is self-financing, by [Proposition 12.6](#),

$$\beta(T)X = \tilde{V}_\varphi(T) = V_\varphi(0) + \sum_{\tau=1}^T \varphi(\tau) \cdot \Delta \tilde{S}(\tau). \quad (69)$$

We know by the no-arbitrage theorem (see [Theorem 12.1](#)) that an equivalent martingale measure $\mathbb{P}^*$ exists; we have to prove uniqueness. So, let $\mathbb{P}_1, \mathbb{P}_2$ be two such equivalent

martingale measures. For $i = 1, 2$, $(\tilde{V}_\varphi(t))_{t=0}^T$ is a $\mathbb{P}_i$ -martingale. So,

$$\mathbb{E}_i(\tilde{V}_\varphi(T)) = \mathbb{E}_i(\tilde{V}_\varphi(0)) = V_\varphi(0), \quad (70)$$

since the value at time zero is non-random ($\mathcal{F}_0 = \{\emptyset, \Omega\}$) and $\beta(0) = 1$. So

$$\mathbb{E}_1(\beta(T)X) = \mathbb{E}_2(\beta(T)X). \quad (71)$$

Since X is arbitrary, $\mathbb{E}_1, \mathbb{E}_2$ have to agree on integrating all non-negative integrands. Taking negatives and using linearity: they have to agree on non-positive integrands also. Splitting an arbitrary integrand into its positive and negative parts: they have to agree on all integrands. Now $\mathbb{E}_i$ is expectation (i.e. integration) with respect to the measure $\mathbb{P}_i$, and measures that agree on integrating all integrands must coincide. So $\mathbb{P}_1 = \mathbb{P}_2$, giving uniqueness as required.

' $\Leftarrow$ ': Assume that the arbitrage-free market $\mathcal{M}$ is incomplete: then there exists a non-attainable $\mathcal{F}_T$ -measurable random variable $X \geq 0$ (a contingent claim). We may confine attention to self-financing strategies φ (which will in the replicating case be automatically admissible). By [Proposition 12.5](#), we may confine attention to the risk assets $S_1, \dots, S_d$, as these suffice to tell us how to handle the numéraire S_0 .

Consider the following set of random variables

$$\tilde{V} := \left\{ Y \in \mathcal{X} : Y = Y_0 + \sum_{t=1}^T \varphi(t) \cdot \Delta \tilde{S}(t), Y_0 \in \mathbb{R}, \varphi \text{ predictable} \right\}.$$

(Recall that Y_0 is $\mathcal{F}_0$ -measurable and set $\varphi = ((\varphi_1(t), \dots, \varphi_d(t)))_{t=1}^T$ with predictable components.) Then by the above reasoning, the discounted value $\beta(T)X$ does not belong to $\tilde{V}$, so $\tilde{V}$ is a *proper* subset of the set $\mathcal{X}$ of all random variables on Ω (which may be identified with $\mathbb{R}^{|\Omega|}$). Let $\mathbb{P}^*$ be a probability measure equivalent to $\mathbb{P}$ under which discounted prices are martingales (such $\mathbb{P}^*$ exist by the no-arbitrage theorem, [Theorem 12.1](#)). Define the scalar product

$$(Z, Y) \rightarrow \mathbb{E}^*(ZY)$$

on random variables on Ω . Since $\tilde{V}$ is a proper subset, there exists a non-zero random variable Z orthogonal to $\tilde{V}$ (since Ω is finite, $\mathbb{R}^{|\Omega|}$ is Euclidean: this is just Euclidean geometry). That is,

$$\mathbb{E}^*(ZY) = 0, \quad \forall Y \in \tilde{V}.$$

Choosing the special $Y = 1 \in \tilde{V}$ given by $\varphi_i(t) = 0, t = 1, 2, \dots, T; i = 1, \dots, d$ and $Y_0 = 1$ we find

$$\mathbb{E}^*(Z) = 0.$$

Write $\|X\|_\infty := \sup\{|X(\omega)| : \omega \in \Omega\}$, and define $\mathbb{P}^{**}$ by

$$\mathbb{P}^{**}(\{\omega\}) = \left(1 + \frac{Z(\omega)}{2\|Z\|_\infty}\right) \mathbb{P}^*(\{\omega\}).$$

By construction, $\mathbb{P}^{**}$ is equivalent to $\mathbb{P}^*$ (same null sets - actually, as $\mathbb{P}^* \sim \mathbb{P}$ and $\mathbb{P}$ has no non-empty null sets, neither do $\mathbb{P}^*, \mathbb{P}^{**}$). As Z is non-zero, $\mathbb{P}^{**}$ and $\mathbb{P}^*$ are *different*. Now

$$\begin{aligned} \mathbb{E}^{**}\left(\sum_{t=1}^T \varphi(t) \cdot \Delta \tilde{S}(t)\right) &= \sum_{\omega \in \Omega} \mathbb{P}^{**}(\omega) \left(\sum_{t=1}^T \varphi(t, \omega) \cdot \Delta \tilde{S}(t, \omega)\right) \\ &= \sum_{\omega \in \Omega} \left(1 + \frac{Z(\omega)}{2\|Z\|_\infty}\right) \mathbb{P}^*(\omega) \left(\sum_{t=1}^T \varphi(t, \omega) \cdot \Delta \tilde{S}(t, \omega)\right). \end{aligned}$$

The ‘1’ term on the right gives

$$\mathbb{E}^*\left(\sum_{t=1}^T \varphi(t) \cdot \Delta \tilde{S}(t)\right),$$

which is zero since this is a martingale transform of the $\mathbb{P}^*$ -martingale $\tilde{S}(t)$ (recall martingale transforms are by definition null at zero). The ‘ Z ’ term gives a multiple of the inner product

$$\left(Z, \sum_{t=1}^T \varphi(t) \cdot \Delta \tilde{S}(t)\right),$$

which is zero as Z is orthogonal to $\tilde{V}$ and $\sum_{t=1}^T \varphi(t) \cdot \Delta \tilde{S}(t) \in \tilde{V}$. By the martingale transform lemma (Lemma 3.4.1), $\tilde{S}(t)$ is a $\mathbb{P}^{**}$ -martingale since φ is an arbitrary predictable process.

Thus $\mathbb{P}^{**}$ is a second equivalent martingale measure, different from $\mathbb{P}^*$. So incompleteness implies non-uniqueness of equivalent martingale measures, as required. $\square$

12.6 Fundamental Theorem of Asset Pricing

We call a measure $\mathbb{P}^*$ under which discounted prices $\tilde{S}(t)$ are $\mathbb{P}^*$ -martingales a *martingale measure*. Such a $\mathbb{P}^*$ equivalent to the actual probability measure P is called an *equivalent martingale measure*.

Theorem 12.3: Fundamental Theorem of Asset Pricing

In an arbitrage-free complete market, there exists a unique martingale measure $\mathbb{P}^$.*

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Monte Carlo Methods

SIMULATION techniques are invaluable in managing complex financial scenarios that involve path-dependent payoffs or distributions without analytical expressions. One prominent application is pricing derivative securities, estimating value at risk, and emulating hedging strategies. Monte Carlo simulation, in particular, proves highly effective for such tasks, offering a blend of simplicity and adaptability as its primary advantages. However, the method's drawback lies in its considerable computational demands, which can be mitigated to some extent. Our approach employs Monte Carlo simulation to generate

simulated option price values, which are then compared against outcomes derived from analytical formulas.

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Whereas a European option can be exercised only at a fixed date, an American option can be exercised any time up to its expiration. The value of an American option is the value achieved by exercising optimally. Finding this value entails finding the optimal exercise rule by solving an optimal stopping problem and computing the expected discounted payoff of the option under this rule. The embedded optimisation problem makes this a difficult problem for simulation.

27.1 Approximate Continuation Values

Regression-based methods posit an expression for the continuation value of the form

$$\mathbb{E}[V_{i+1}(X_{i+1}) \mid X_i = x] = \sum_{r=1}^M \beta_{ir} \psi_r(x), \quad (72)$$

for some basis functions $\psi_r : \mathbb{R}^d \rightarrow \mathbb{R}$ and constants $\beta_{ir}, r = 1, \dots, M$. Using the notation C_i for the continuation value at time i , we may equivalently write eq. (72) as

$$C_i(x) = \beta_i^\top \psi(x), \quad (73)$$

with

$$\beta_i^\top = (\beta_{i1}, \dots, \beta_{iM}), \quad \psi(x) = (\psi_1(x), \dots, \psi_M(x))^\top.$$

One could use different basis functions at different exercise dates, but to simplify notation, we suppress any dependence of ψ on i .

Assuming a relation of the form eq. (72) holds, the vector β_i is given by

$$\beta_i = (\mathbb{E}[\psi(X_i)\psi(X_i)^\top])^{-1} \mathbb{E}[\psi(X_i)V_{i+1}(X_{i+1})] \equiv B_\psi^{-1} B_{\psi V}. \quad (74)$$

Here, B_ψ is the indicated $M \times M$ matrix (assumed nonsingular) and $B_{\psi V}$ the indicated vector of length M . The variables (X_i, X_{i+1}) inside the expectations have a joint distribution corresponding to the state of the underlying Markov chain at dates i and $i + 1$.

The coefficients β_{ir} could be estimated from observations of pairs $(X_{ij}, V_{i+1}(X_{i+1,j}))$, $j = 1, \dots, b$, each consisting of the state at time i and the corresponding option value at time $i + 1$. Consider, in particular, independent paths $(X_{1j}, \dots, X_{mj})$, $j = 1, \dots, b$, and suppose for a moment that the values $V_{i+1}(X_{i+1,j})$ are known. The least-squares estimate of β_i is then given by

$$\hat{\beta}_i = \hat{B}_\psi^{-1} \hat{B}_{\psi V},$$

where $\hat{B}_\psi$ and $\hat{B}_{\psi V}$ are the sample counterparts of B_ψ and $B_{\psi V}$. More explicitly, $\hat{B}_\psi$ is the $M \times M$ matrix with qr entry

$$\frac{1}{b} \sum_{j=1}^b \psi_q(X_{ij}) \psi_r(X_{ij})$$

and $\hat{B}_{\psi V}$ is the M -vector with r th entry

$$\frac{1}{b} \sum_{k=1}^b \psi_r(X_{ik}) V_{i+1}(X_{i+1,k}). \quad (75)$$

All of these quantities can be calculated from function values at pairs of consecutive nodes $(X_{ij}, X_{i+1,j})$, $j = 1, \dots, b$. In practice, V_{i+1} is unknown and must be replaced by estimated values $\hat{V}_{i+1}$ at downstream nodes. The estimate $\hat{\beta}_i$ then defines an estimate

$$\hat{C}_i(x) = \hat{\beta}_i^\top \psi(x), \quad (76)$$

of the continuation value at an arbitrary point x in the state space $\mathbb{R}^d$. This, in turn, defines a procedure for estimating the option value.

Algorithm 1 Regression-Based Pricing Algorithm

- 1: Simulate b independent paths $\{X_{1j}, \dots, X_{mj}\}$, $j = 1, \dots, b$, of the Markov chain.
- 2: At terminal nodes, set $\hat{V}_{mj} = h_m(X_{mj})$ for $j = 1, \dots, b$.
- 3: **for** $i = m - 1$ **down to** 1 **do**
- 4: Given estimated values $\hat{V}_{i+1,j}$, $j = 1, \dots, b$, use regression to calculate $\hat{\beta}_i = \hat{B}_\psi^{-1} \hat{B}_{\psi V}$.
- 5: Compute estimated continuation values $\hat{C}_i(X_{ij})$ using [eq. \(76\)](#).
- 6: **for** $j = 1, \dots, b$ **do**
- 7: Set value based on immediate exercise versus estimated continuation:

$$\hat{V}_{ij} = \max \{h_i(X_{ij}), \hat{C}_i(X_{ij})\}. \quad (77)$$

- 8: **end for**
 - 9: **end for**
 - 10: Set $\hat{V}_0 = (\hat{V}_{11} + \dots + \hat{V}_{1b})/b$.
-

Definition 27.1: American Contingent Claim

An *American contingent claim* $X^a = (X, \mathcal{T}_{[0,T]})$ expiring at time T consists of a sequence of payoffs $(X_t)_{t=0}^T$, where X_t is an $\mathcal{F}_t$ -measurable random variable for $t = 0, 1, \dots, T$, and of the set $\mathcal{T}_{[0,T]}$ of admissible exercise policies.

Proposition 27.1: Arbitrage Price of American Put

The arbitrage price P_t^a of an American put option equals

$$P_t^a = \max_{\tau \in \mathcal{T}_{[t,T]}} \mathbb{E}_{\mathbb{P}^*} \left(\hat{r}^{-(\tau-t)} (K - S_\tau)^+ \mid \mathcal{F}_t \right), \quad \forall t \leq T. \quad (78)$$

Moreover, for any $t \leq T$ the stopping time τ_t^* that realizes the maximum in eq. (78) is given by the expression (by convention $\min \emptyset = T$)

$$\tau_t^* = \min \left\{ u \in \{t, t+1, \dots, T\} \mid (K - S_u)^+ \geq P_u^a \right\}. \quad (79)$$

Proposition 27.2: Arbitrage Price of American Claim in CRR

For every $t \leq T$, the arbitrage price $\pi(X^a)$ of an arbitrary American claim X^a in the CRR model equals

$$\pi_t(X^a) = \max_{\tau \in \mathcal{T}_{[t,T]}} \mathbb{E}_{\mathbb{P}^*} \left(\hat{r}^{-(\tau-t)} X_\tau \mid \mathcal{F}_t \right). \quad (80)$$

The price process $\pi(X^a)$ can be determined using the following recurrence relation, for every $t \leq T-1$,

$$\pi_t(X^a) = \max \left\{ X_t, \mathbb{E}_{\mathbb{P}^*} \left(\hat{r}^{-1} \pi_{t+1}(X^a) \mid \mathcal{F}_t \right) \right\} \quad (81)$$

subject to the terminal condition $\pi_T(X^a) = X_T$. In the case of a path-independent American claim X^a with the reward function h we have that, for every $t \leq T-1$,

$$\pi_t(X^a) = \max \left\{ h(S_t, t), \hat{r}^{-1} \left(p_* \pi_{t+1}^u(X^a) + (1 - p_*) \pi_{t+1}^d(X^a) \right) \right\}, \quad (82)$$

where, for a generic stock price value S_t at time t , we write $\pi_{t+1}^u(X^a)$ and $\pi_{t+1}^d(X^a)$ to denote the values of the price process $\pi(X^a)$ at time $t+1$ in the nodes that correspond to the upward and downward movements of the stock price during the time-period $[t, t+1]$, that is, for the values uS_t and dS_t of the stock price at time $t+1$, respectively.

28 Lookback Options

29 Asian Options

An *Asian option* (or *average option*) defines a class of options, whether European or American style, whose terminal payoff depends on the average asset value over a specific period within the option's life. Due to this averaging feature, Asian options are particularly suitable for thinly traded assets or commodities. Unlike standard options, they are typically less expensive and offer greater robustness against price manipulation near the expiry date.

Asian options were first traded in Tokyo in 1987, and have become particularly popular in commodities trading.

Let T denote the exercise date, and let $0 \leq T_0 < T$ represent the commencement of the averaging period. The payoff at expiry for an Asian call option is defined as

$$C_T^A := (A_S(T_0, T) - K)^+, \quad (83)$$

where

$$A_S(T_0, T) = \frac{1}{T - T_0} \int_{T_0}^T S_u \, du \quad (84)$$

represents the arithmetic average of the asset price over the time interval $[T_0, T]$. Here, K denotes the fixed strike price, and the asset price S is assumed to follow a geometric Brownian motion.

The main difficulty in pricing and hedging Asian options is due to the fact that the random variable $A(T_0, T)$ does not have a lognormal distribution. This feature makes the task of finding an explicit formula for the price of an Asian option surprisingly involved.

For this reason, early studies of Asian options were based either on approximations or on the direct application of the Monte Carlo method. The numerical approach to the valuation of Asian options, is based on the approximation of the arithmetic average using the geometric average. Note first that it is natural to substitute for the continuous-time

average $A(T_0, T)$ its discrete-time counterpart

$$A_S^n(T_0, T) = \frac{1}{n} \sum_{i=0}^{n-1} S_{T_i}, \quad (85)$$

where $T_i = T_0 + i(T - T_0)/n$. Furthermore, the arithmetic average $A_S^n(T_0, T)$ can be replaced with the geometric average, denoted by $G_S^n(T_0, T)$, in what follows. Recall that the random variables S_{T_i} , $i = 1, \dots, n$ are given explicitly by the expression

$$S_{T_i} = S_{T_0} \exp \left(\sigma(W_{T_i}^* - W_{T_0}^*) + \left(r - \frac{1}{2}\sigma^2 \right) (T_i - T_0) \right), \quad (86)$$

where W^* is a standard Brownian motion under the martingale measure $\mathbb{P}^*$. Hence the geometric average admits the following representation

$$G_S^n(T_0, T) = \left(\prod_{i=0}^{n-1} S_{T_i} \right)^{1/n} = c S_{T_0} \exp \left(\frac{\sigma}{n} \sum_{i=0}^{n-1} (n-i-1) (W_{T_{i+1}}^* - W_{T_i}^*) \right) \quad (87)$$

for a strictly positive constant c . In view of the independence of increments of the Brownian motion, the last formula makes clear that the geometric average $G_S^n(T_0, T)$ has a lognormal distribution under $\mathbb{P}^*$. The approximate Black-Scholes-like formula for the price of an Asian call option can thus be easily found by the direct evaluation of the conditional expectation

$$\tilde{C}_{T_0}^n = e^{-r(T-T_0)} \mathbb{E}_{\mathbb{P}^*} \left((G_S^n(T_0, T) - K)^+ \mid \mathcal{F}_{T_0} \right). \quad (88)$$

It appears, however, that such an approach significantly underprices Asian call options. To overcome this deficiency, one may directly approximate the true distribution of the arithmetic average using an approximate distribution, typically a lognormal distribution with the appropriate parameters

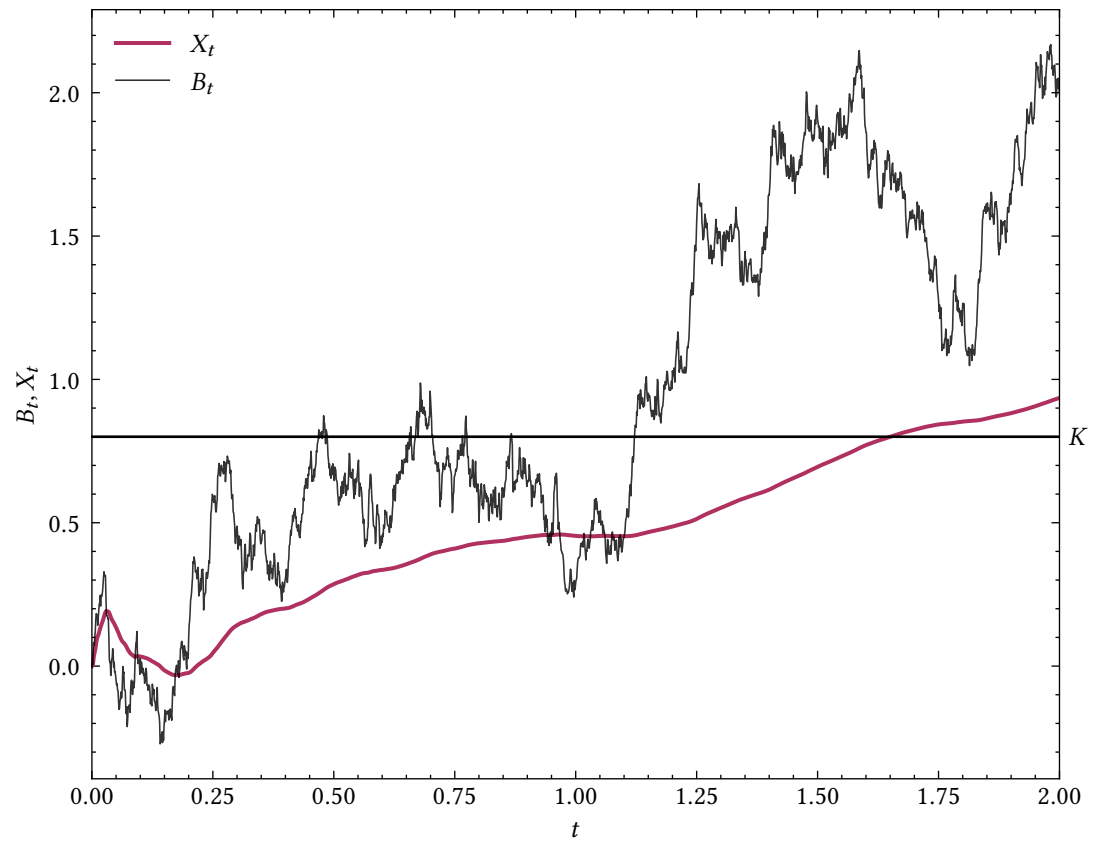


Figure 5: Brownian motion B_t and its moving average X_t

30 Basket Options

31 Quantile Options

Part IX

Hedging

32 Delta Hedging

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Appendices

A Existence of Lebesgue Measure

B Linear Algebra and Functional Analysis

C Multivariate Differentiation

D Probability and Statistics

E Python Primer